

Muhammad Zain

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EDUCATION

University of Calgary

Bachelor of Science in Software Engineering

Calgary, AB

September 2022 - April 2027

- **Relevant Coursework:** Software Architecture, Software Testing and Quality, Software Performance Evaluation, Computational Data Management, Concurrent and Distributed Systems, Algorithmic Design and Analysis

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS, RISC-V Assembly
- **Application Development:** React, Node.js, Next.js, Flask, REST APIs, Client-Server Architecture, Responsive UI
- **Data, Testing and Storage:** MySQL, PostgreSQL, SQLite, Cypress, JUnit 5, Mockito, Jest, Locust, Query Planning
- **Engineering Practices and Tooling:** Git, GitHub, Docker, Docker Compose, GCP, CMake, Qt, Swing, Agile Workflows

EXPERIENCE

Enactus - Paragon

Software Development Intern

Calgary, AB

September 2025 - May 2026

- Designed a **spatiotemporal data pipeline** aligning satellite, weather, and event datasets onto a canonical **30 km grid** for weekly wildfire risk analysis.
- Implemented **leakage-safe temporal validation** and grid-based joins, preventing spatial and time contamination across multi-year **machine learning** experiments.
- Engineered feature workflows for **extreme class imbalance**, evaluating gradient-boosted models with **PR-AUC** and Recall@Top-K to prioritize high-risk regions.
- Standardized reproducible outputs including **Parquet tables**, model artifacts, and **interactive risk maps** for engineering handoff and operational review.

PROJECTS

AI Resume and Career Coach 🔄

React, Flask, MySQL, Docker, Gemini API, REST APIs

Calgary, AB

January 2026 - April 2026

- Built a **full-stack web app** for resume uploads, job analysis, history tracking, and interview coaching workflows.
- Integrated a **hybrid token-overlap + LLM pipeline** using structured JSON prompts to score resume-to-job fit.
- Developed **Flask REST APIs** with MySQL connection pooling, file parsing, authentication, and persisted analysis results.
- Containerized frontend, backend, and database services with **Docker Compose** for repeatable local deployment.

ProfileHub Performance Evaluation 🔄

Python, Flask, React, TypeScript, MySQL, Locust, GCP

Calgary, AB

January 2026 - April 2026

- Built and deployed a **Flask + React** profile app on Google Cloud Run with Cloud SQL and GCS.
- Designed a **72-run Locust experiment** across workloads, payload sizes, concurrency levels, and repeated performance trials.
- Developed **Python analysis scripts** computing 95% confidence intervals for latency, p95, throughput, and failure rates.
- Diagnosed MySQL connection-pool leaks under load, improving **database reliability** during concurrent user simulations.

CityX Subway Display System 🔄

Java, Swing, MySQL, REST APIs, Multithreading

Calgary, AB

June 2025 - July 2025

- Engineered a **real-time subway display** with animated tracking, rotating ads, and live API-driven content.
- Structured a **MySQL ad scheduler** with indexed rules, cutting query latency by **35%**.
- Integrated weather and news APIs with **caching layers** and retry logic, maintaining **90% reliability**.
- Designed a modular Swing interface with **multithreaded pipelines**, enabling smooth ad-to-map transitions.

Flight Operations Manager 🔄

C++, Qt, SQLite/CSV, Data Visualization

Calgary, AB

February 2025 - March 2025

- Developed a **Qt desktop application** to manage flights, passengers, schedules, and **500+ operational records**.
- Created an **interactive seat map** with validation layers and conflict checks, reducing booking errors by **40%**.
- Optimized SQLite storage and query paths for larger datasets, achieving **~35% faster** execution times.
- Enhanced system stability by debugging **segmentation faults** and error paths, sustaining **~100% uptime**.